magnetic needle is exactly horizontal and also coil (mounted on the frame) is exactly vertical.

3. The plane of the coil is kept parallel to the small magnetic needle by rotating the coil about its vertical axis. So, the coil remains in magnetic meridian.



Figure 3.23 Compass box

4. The compass box (as shown in Figure 3.23) is rotated such that the pointer reads $0^o - 0^o$

Theory

The circuit connection for Tangent Galvanometer (TG) experiment is shown in Figure 3.24. When no current is passed through the coil, the small magnetic needle lies along horizontal component of Earth's magnetic field. When the circuit is switched ON, the electric current will pass through the circular coil and produce magnetic field. The magnetic field produced due to the circulatory electric current is discussed **(in section 3.8.3).** Now there are two fields which are acting mutually perpendicular to each other. They are:

(1) the magnetic field (B) due to the electric current in the coil acting normal to the plane of the coil.

(2) the horizontal component of Earth's magnetic field (B_H)

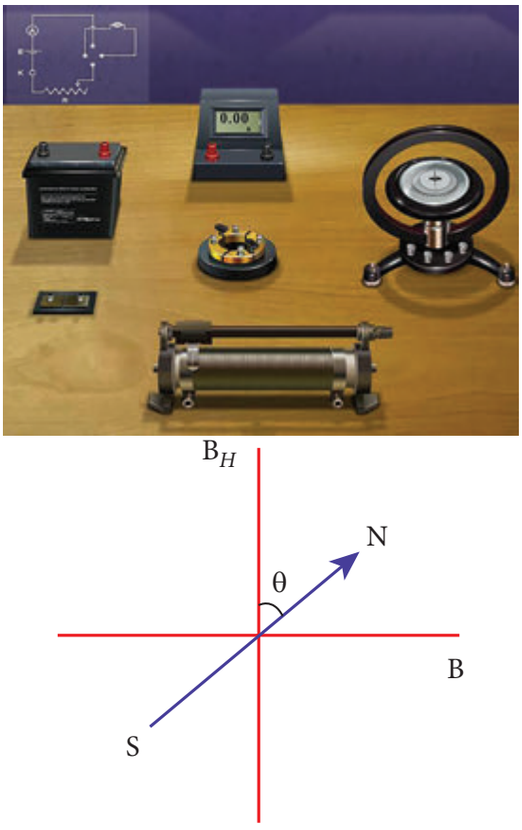


Figure 3.24 (a) circuit connection (b) resultant position of pivoted needle

Because of these crossed fields, the pivoted magnetic needle deflects through an angle θ. From tangent law (equation 3.29),

$$B = B_H \tan\theta$$

When an electric current is passed through a circular coil of radius R having N turns, the magnitude of magnetic field at the center is

$$B = \mu_\circ \frac{NI}{2R} \qquad (3.30)$$

From equation (3.29) and equation (3.30), we get

$$\mu_\circ \frac{NI}{2R} = B_H \tan\theta$$

The horizontal component of Earth's magnetic field can be determined as

$$B_H = \mu_\circ \frac{NI}{2R}\frac{1}{\tan\theta} \text{ in tesla} \qquad (3.31)$$

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